

privileged information (many of these are hackers in need of attention and self-respect—some are harmless, but some can cause harm); or employees who attack their own companies as a form of revenge. CERN defines this threat as, “A malicious insider threat to an organisation is a current or former employee, contractor or other business partner who has or had authorised access to an organisation’s network, system or data and intentionally exceeded or misused that access in a manner that negatively affected the confidentiality, integrity, or availability of the organisation’s information or information systems.”

Once the insiders get access to the sensitive information, they can use it for their own personal needs through spoofing, tampering, information disclosure, etc. Sometimes the insiders may already have permission to access data at many internal levels (depending on the designation), which makes the situation easier for them. Systems that depend solely on the Cloud Service Provider (CSP) for security are at great risk here. To address this, the following measures can be taken: specifying human resource requirements as part of legal contracts, existence of transparency in the overall information security and management practices along with compliance reporting, setting up insider-theft prevention programs and determining security breach notification processes.

Abuse of cloud services

The services and resources rented out by the cloud service provider can be accessed by the attackers for illicit purposes. For instance, it might take attackers ages to decrypt the data stolen with their own limited resources but using the cloud computing services, it could take only minutes to crack. The cyber criminal may also use the cloud services to stage attacks like DDoS, DoS, severe malware damage or distribute pirated software. In 2013, a group of cyber criminals had used Google Cloud Messaging as a command and control (C&C) server for their malware that was designed to send SMS messages to premium rate numbers, steal messages and contacts, and display shady advertisements, by responding to which they might be led to other malicious elements.

There is a great need to confront this threat, and some simple steps could include following the principle of ‘trust-but-verify’, centralised log monitoring for users with limited privileges, setting alerts for log policy violations or unauthorised access to encrypted data, as well as encryption key breach alerts that assist in early detection of user abuse. Security Information and Event Management (SIEM) solutions that are currently available can also be configured to alert admins about harmful activities on the cloud. The administrators must use tools to monitor CPU, memory, network and disk usage in order to detect abusive usage.

Shared technology vulnerabilities

Sharing technology can create vulnerabilities on the cloud. Various cloud services are delivered to customers through shared infrastructures, platforms and applications. Hence, the threat of shared vulnerabilities exists on all cloud delivery models.

A compromise in an integral piece of shared technology could expose more than one customer. In fact, it could expose the entire environment to a potential breach, making the cloud insecure. A single vulnerability or misconfiguration can lead to a compromise across the entire cloud. Security best practices like proper installation and configuration, monitoring the environment for unauthorised changes in activities, promoting strong authentication and access procedures, etc, should be followed to confront this threat.

Cloud security controls

There are many types of cloud security control measures to tackle these threats, which fall under the following categories.

Deterrent controls

These control measures use the straightforward way of displaying the message ‘DANGER’ if you tamper with the system unnecessarily. They warn the attackers by highlighting the fact that they could face adverse consequences in case of malpractices. Examples include a log on warning, a prompt warning on the screen when users try to access unauthorised information or page, creating appropriate access banners on entry pages, stopping the malicious activity at its initial stage itself, etc.

Preventive controls

Preventive controls strengthen the system against incidents by reducing the vulnerabilities, if not completely eradicating them. As an example, unauthorised intruders can be locked out by providing strong authentication for all cloud users; people without authentication will then not be able to access sensitive information.

Detective controls

In the event of a threatening incident, detective control measures are intended to identify and characterise an incident in progress, and to address the preventive or corrective controls to look into the situation and handle it. An example of this is sounding the intruder alarm and alerting the security guards or police.

Corrective controls

Corrective controls are intended to limit the extent of any damage caused by an incident. They come into effect during or after an incident. Restoring system back-ups in order to rebuild a compromised system is an example of a corrective control. 

References

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